Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Wearable IoT–Based Safety and Productivity Intelligence Platform for Workforce-Intensive Industries			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Mechanical · Materials		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	Measurement / Sensor Technologies for Automation	Electrical and Electronic Components	Sensor Components
Concept of Proposed Technology	The proposed technology aims to develop a wearable IoT–based safety and productivity intelligence platform for workforce-intensive industries. The platform enables real-time monitoring of worker safety states, spatial context, and task-level productivity by transforming wearable and location-based sensing data into actionable insights for operational decision-making.			

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	The research institute has developed and validated initial laboratory-scale wearable IoT prototypes and core motion recognition algorithms at approximately TRL 3, providing a proof-of-concept foundation for the proposed joint R&D effort. The system integrates wearable IoT sensing devices embedded in Smart Personal Protection Equipment (PPE), Bluetooth Low Energy (BLE)–based localization infrastructure, and AI-driven analytics within a unified platform architecture. Wearable sensors capture worker motion and posture data, while BLE beacons deployed across worksites provide worker and equipment localization, zone awareness, and proximity detection. These data streams are fused in real time to support safety alerting and productivity analytics at both individual and workforce levels. The platform supports detection of safety-critical events such as unsafe posture, hazardous zone entry, and unsafe proximity to mobile equipment, while also enabling task-level productivity analysis through hierarchical motion–activity–task recognition models. Construction and manufacturing environments serve as the primary application domains due to their labor intensity, safety risks, and operational complexity; however, the platform architecture is designed to be transferable to other workforce-intensive industries with minimal reconfiguration.
Rationale for Joint R&D Project	**The main objective of this project** is to develop and field-validate a wearable IoT–based safety and productivity intelligence platform for workforce-intensive industries, advancing the system to TRL 8, with primary applications in construction and manufacturing. Achieving this objective requires translating validated wearable sensing and AI-based analytics technologies, proven at the laboratory scale, into a deployment-ready platform capable of sustained operation in real-world conditions. System-level validation in operational environments necessitates close collaboration between a research institute and an industrial partner, as both advanced analytics development and field-oriented system integration capabilities are essential. The research institute contributes expertise in wearable sensing, hierarchical AI-based activity and task recognition, and safety and productivity analytics, which form the platform's core intelligence. The Korean SME partner provides complementary capabilities in wearable hardware integration, system engineering, and field deployment, including the development of compact, user-friendly wearable devices suitable for practical field operation**,** enabling translation of research outcomes into a deployment-ready system validated in real environments. Joint R&D enables coordinated co-design of hardware, software, and system architecture under practical constraints such as durability, usability, wireless reliability, and operational robustness. This collaboration also ensures alignment of technical development with deployment requirements, certification readiness, and post-project commercialization planning.

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Key Words	1. IoT-based wearable system 2. Smart PPE 3. BLE Localization 4. Motion & Posture Safety Analytics 5. Motion-Activity-Task Recognition 6. Workforce Productivity Analytics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (4)	**End of the Project**	Level (8)
R&D Objectives	The objectives of the proposed R&D project are as follows: 1. To develop deployment-ready wearable IoT devices integrated into Smart PPE, including optimization of sensor configuration for reliable motion capture and user compliance. 2. To establish robust worker and equipment localization and proximity sensing using BLE infrastructure under dynamic operational conditions common to workforce-intensive industries. 3. To develop AI-based hierarchical recognition models that infer task-level operations and productivity metrics, enabling workforce-level productivity intelligence (e.g., bricklaying at approximately 60 bricks per hour and slab concrete placement at approximately 1.5 m^3 per worker-hour). 4. To implement real-time safety functions, including unsafe posture detection, unsafe proximity detection, hazard-zone alerts, and safety event detection. 5. To integrate sensing, analytics, alerting, and visualization into a unified intelligence platform supporting site-level safety oversight and productivity management.			
R&D Target Specifications	**Motion-level recognition accuracy (≥ 80%)** The system shall reliably recognize fundamental worker motions that serve as inputs for higher-level activity and task inference. **Activity- and task-level recognition reliability (≥ 80% or F1 ≥ 0.80)** The system shall provide stable task-level recognition performance suitable for productivity analysis under field conditions. **Proximity hazard detection accuracy (≥ 85% or F1 ≥ 0.85)** The system shall accurately detect hazardous proximity events between workers, equipment, and restricted zones while limiting excessive false alerts.			

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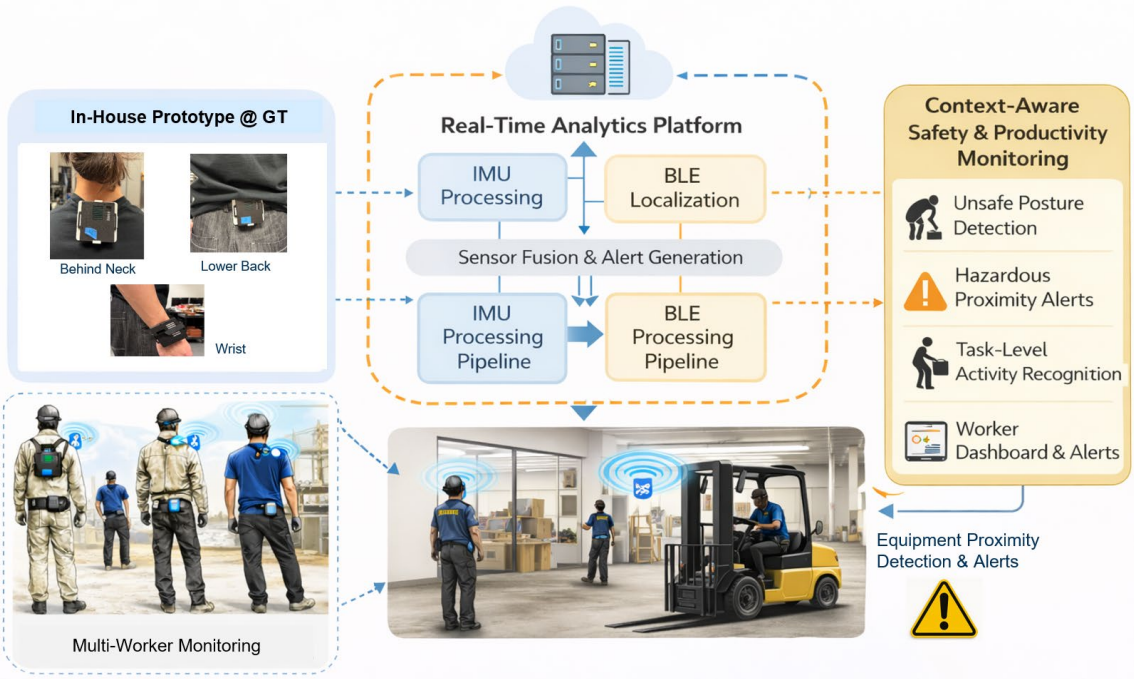
	Safety event detection recall (≥ 85%) The system shall prioritize recall to minimize missed safety-critical events during operation. **Beacon-based localization accuracy (≤ 3 m or ≥ 80% zone-level accuracy)** The localization system shall provide sufficient spatial resolution for proximity detection, zone monitoring, and trajectory analysis. **Alert response latency (≤ 1 s)** The system shall deliver safety alerts within one second to support timely hazard response. **System availability during operation (≥ 95%)** The platform shall maintain high availability to support continuous monitoring during active operations. **Hardware durability and protection (≥ IP65)** Wearable devices shall be protected against dust and water exposure typical of workforce-intensive industry environments.
R&D Plan	This R&D project will be carried out over a three-year period to advance the proposed IoT-based wearable workforce intelligence system toward a deployable platform for industrial applications. The system will be primarily evaluated in complex, dynamic, and challenging environments such as construction sites, while also considering its applicability to other industrial settings, including manufacturing and warehouse facilities. Fig. 1 shows representative laboratory-scale PPU prototypes developed by the research institute (TRL 3) and the overall framework of the proposed wearable IoT system.  ***Fig. 1. Multi-Worker Wearable IoT Framework for Real-time Safety & Productivity Monitoring***

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Year	Focus	Key Activities and Outputs
Year 1	System architecture and core integration	○ Defined system requirements and representative workflows ○ Integrated sensing and platform architecture (IMU + BLE + analytics) ○ Baseline motion recognition and localization algorithms ○ Initial performance targets and validation framework **(TRL 4-5)**
Year 2	System refinement and controlled field validation	○ Field-ready compact PPE hardware prototype ○ Activity- and task-level AI analytics implemented ○ Integrated safety and productivity monitoring platform ○ Pilot field validation results and iterative refinements **(TRL 6–7)**
Year 3	Extended field validation and TRL 8 demonstration	○ Extended field deployment in active sites ○ Validated system performance against target specifications ○ Proven usability and operational stability ○ Deployment guidelines and certification-ready documentation **(TRL 8)**

[Year 1] System Architecture Design and Core Technology Development

(1.1) System Definition and Sensing Architecture Design

- Refine application scenarios, system requirements, and representative workflows based on existing TRL-3 wearable sensing prototypes, in collaboration with industry stakeholders, focusing on key construction trades (e.g., masonry, HVAC, painting) and realistic site conditions.
- Identify and prioritize critical safety scenarios and operational constraints relevant to workforce-intensive environments.
- Refine the integrated sensing and system architecture combining wearable IMU sensors, BLE-based localization, and a central analytics platform, including data flow, communication, and user interface design.
- Update performance targets for motion sensing, localization accuracy, and analytics reliability to support progression from laboratory validation to system-level evaluation.

(1.2) Core Sensing and Algorithm Development

- Refine wearable sensing and localization configurations by evaluating minimal sensor setups (e.g., 2–3 IMUs) that balance recognition performance, comfort, and safety-gear compliance, building on existing prototypes.
- Refine BLE beacon deployment strategies considering site geometry, dynamic layouts, and wireless interference, and establish practical localization coverage and accuracy targets.
- Enhance baseline motion-level recognition algorithms using wearable IMU data and integrate improved BLE-based localization to provide spatial context for safety and productivity analytics.

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	[Year 2] Productization and Integrated System Validation **(2.1) Smart PPE Productization and Algorithm Expansion** • Advance Smart PPE hardware toward field-ready maturity by refining compact form-factor design, durability, battery life, ergonomics, power management, and manufacturability based on pilot feedback. • Expand AI algorithms from motion recognition to activity- and task-level inference using hierarchical models tailored to construction workflows. • Improve algorithm robustness to sensor noise, worker variability, and environmental disturbances encountered in real sites. **(2.2) Integrated Platform Development and Pilot Validation** • Develop an end-to-end platform integrating real-time analytics, safety alerting (audio/vibration), visualization, and basic device management. • Conduct pilot-scale field testing at construction sites to evaluate recognition accuracy, safety responsiveness, system reliability, and usability. • Iteratively refine hardware, algorithms, and platform components based on pilot data, performance evaluation, and user feedback. **[Year 3] Extended Field Validation and Deployment Readiness (TRL 8)** **(3.1) Large-Scale Field Deployment and Performance Evaluation** • Deploy the fully integrated system at active construction sites for extended operation under diverse work conditions, schedules, and environments. • Evaluate system performance against final target specifications, including accuracy, latency, localization performance, availability, and alert effectiveness. • Validate practical usability and adoption readiness for both workers and site managers, identifying training and operational considerations for broader deployment. **(3.2) Deployment Readiness, Certification Preparation, and Cross-Industry Applicability** • Refine deployment procedures, maintenance workflows, and operational guidelines to support repeatable, multi-site adoption. • Prepare technical documentation, validation reports, and performance summaries required for certification and standardization. • Assess applicability of the wearable sensing and analytics framework to other workforce-intensive industries, such as manufacturing and industrial plants, and identify system components that support cross-industry adaptation. • Advance the integrated system to TRL 8 through system-level validation in operational environments and establish a clear pathway for post-project commercialization in collaboration with the Korean SME partner.

4. R&R (Roles and Responsibilities)

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Expected R&R of Research Institute	The research institute will be responsible for developing the core sensing intelligence, AI algorithms, and analytical frameworks that enable reliable real-time safety monitoring and task-level productivity analytics for workforce-intensive industries, with primary application to construction and manufacturing. These efforts will provide the technical and scientific foundation required for system-level validation, deployment readiness, and post-project commercialization. The specific roles and responsibilities of the research institute include: • Development of IMU-based hierarchical motion, activity, and task recognition algorithms that enable real-time task inference from wearable sensing data under realistic operational conditions. • Development of AI-based safety incident detection methods aligned with worker-related safety incidents defined in the Korean Occupational Safety and Health Act (2023), including fall-related events, struck-by or collision incidents, slip and trip events, caught-in or between accidents, and abnormal motion patterns. • Design of the overall system architecture and spatiotemporal data fusion framework, integrating wearable motion sensing outputs with BLE-based indoor localization and trajectory information to support real-time context-aware analytics. • Development and validation of AI-based safety and productivity analytics, including proximity-based hazard detection, task-level time allocation, workforce distribution metrics, and productivity indicators suitable for field evaluation and managerial decision support. • Establishment of performance evaluation protocols and validation methodologies to support pilot testing and extended field trials in collaboration with the industrial partner. • Provision of technical guidance and collaborative support to the Korean SME partner throughout system integration, pilot deployment, and performance evaluation activities.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for productization, system integration, field deployment, and commercialization-oriented preparation of the proposed wearable IoT–based workforce intelligence platform, enabling reliable real-world operation and transition toward market adoption following TRL 8 validation. The specific roles and responsibilities of the Korean SME partner include: • Design and manufacture deployment-ready wearable hardware units, including compact form-factor devices, enclosures, and industrial design, with emphasis on power management, durability, and protection (e.g., dust and water resistance and impact robustness), as well as design for manufacturability and assembly (DFMA) suitable for workforce-intensive industry environments such as construction and manufacturing. • Deploy and configure IoT system infrastructure for workers and mobile equipment (e.g., forklifts), and integrate wearable devices with BLE-based localization infrastructure to support stable localization, zone awareness, and trajectory monitoring.

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	• Perform system-level integration of wearable devices, BLE infrastructure, the central intelligence platform, and local communication networks (e.g., Wi-Fi, cellular networks, hotspots, or ad-hoc networks), ensuring reliable communication, real-time data transmission, and operational stability in field environments. • Execute pilot deployments and extended field testing, including on-site installation, system operation, data collection, and validation of safety, productivity, reliability, and usability performance under real operational conditions. • Lead customer discovery and stakeholder engagement activities in Korea by working with construction and manufacturing companies, site operators, safety managers, and workers to identify user requirements and validate the practical value of the system, with field-driven feedback informing system refinement. • Contribute to deployment readiness and post-project commercialization preparation, including system refinement, certification and standardization readiness, manufacturing planning, and development of a market entry strategy based on industrial expertise.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project will deliver a **deployment-ready** wearable IoT– and AI-based workforce safety and productivity intelligence platform for workforce-intensive industries, with primary application to construction and manufacturing. The project emphasizes system integration, field validation, and operational readiness rather than full market launch. **Expected Outcomes** • Deployment-ready integration of a wearable IoT monitoring platform supporting continuous on-site operation and centralized safety and productivity oversight. • Field validation of safety performance, including reliable detection of worker-related safety incidents and timely alert delivery under operational conditions. • Validated task-level productivity analytics based on hierarchical motion–activity–task inference, enabling measurement of task-level time allocation and workforce distribution. • Demonstrated system reliability, scalability, and operational readiness through pilot deployments and extended field validation, supporting certification preparation and advancement to TRL 8. **Expected Impact** • From an industrial perspective, the project will provide the Korean SME partner with a field-validated platform, reusable analytics components, and deployment experience that reduce technical and market-entry risks following project completion.

	• The proposed sensing and analytics framework is transferable to other workforce-intensive industries, including manufacturing, logistics, and industrial facilities. • More broadly, the project will contribute to improved worker safety, enhanced workforce utilization, and data-driven operational decision-making, while strengthening the global competitiveness of Korean SMEs in AI-enabled industrial safety and workforce intelligence technologies.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization